Homework #4 - Spaceflight Mechanics

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1 Part 1

1.1 Problem

Consider the following simplification hypotheses:

- circular and coplanar orbits of planets
- negligible residence times in the spheres of influence of the planets

The following table provides the necessary data.

	orbit radius R	planet radius r	solar day d	gravitational parameter μ
Earth (E)	149500000 km	6371 km	86400 s	398600 km^3/s^2
Mars (M)	227800000 km	3397 km	88775 s	42828 km^3/s^2
Sun (S)	-	-	-	1.327×10^{11} km^3/s^2

TABLE 1. Solar system data

First of all, consider a Hohmann transfer between the heliocentric orbits of radius R_M and R_E and evaluate v_∞, the travel time, the position of Mars at departure and the synodic period. Evaluate the angles and times of an Earth-Mars-Earth mission consisting of Hohmann transfers.

1.2 Earth-Mars Hohmann transfer

We need to design first the heliocentric leg of the transfer, then we'll be able to study the problem in detail inside Earth and Mars' soi. A simple Hohmann transfer is here used.

$$a_H = \frac{R_M + R_E}{2} = 1.8865 \times 10^8 \ km$$

$$e_H = \frac{R_M - R_E}{R_M + R_E} = 0.2075$$

$$V_{departure} = \sqrt{\frac{\mu_S}{a_H}} \cdot \sqrt{\frac{1+e_H}{1-e_H}} = 32.74 \ km/s$$

$$V_{\infty,dep} = |V_{departure} - V_E| = 2.95 \ km/s$$

$$V_{arrival} = \sqrt{\frac{\mu_S}{a_H}} \cdot \sqrt{\frac{1-e_H}{1+e_H}} = 21.49 \ km/s$$

$$V_{\infty,arr} = |V_M - V_{arrival}| = 2.65 \ km/s$$

$$TOF = \pi\sqrt{\frac{a_H^3}{\mu_S}} = 258.63 \ days$$

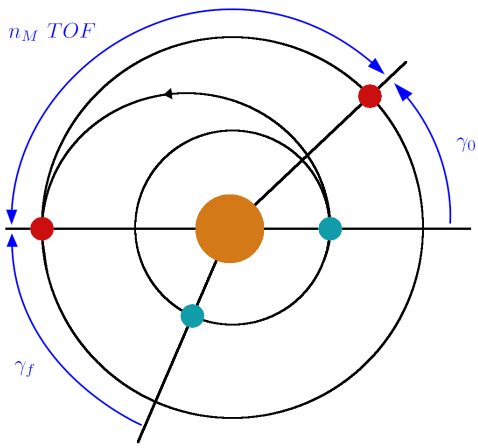
1.3 Angles and time of the mission

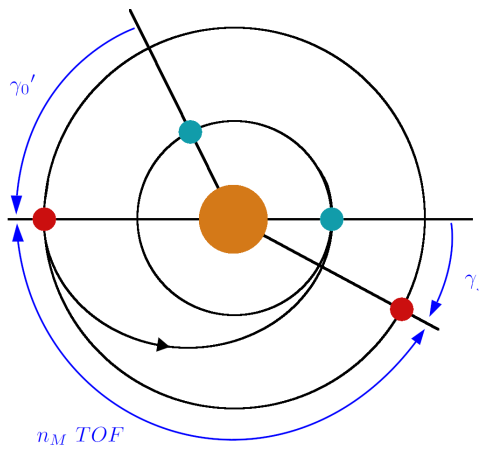
We need to make sure we depart from Earth the moment that ensure us to find Mars on arrival. The angles that Earth and Mars form within the Solar System are essential to estimate the flight time and the time required for the next mission.

$$
\begin{aligned}
&T_{synodic} = \frac{2\pi}{n_M - n_E} = 779.16\ days\\
&\gamma_0 = \pi - n_M\ TOF = 44.35\ deg\\
&\gamma_f = \pi - n_E\ TOF = -75.15\ deg\\
&\gamma_0' = -\gamma_f = 75.15\ deg\\
&t_{wait} = \frac{-2\gamma_f - 2\pi}{n_M - n_E} = 453.86\ days\\
&TOF_{tot} = t_{wait} + 2\ TOF = 971.13\ days = 2.66\ years
\end{aligned}
$$

$$
\begin{cases} \theta_M = \theta_M(0) + n_M t \\ \theta_E = \theta_E(0) + n_E t \end{cases} \quad \Rightarrow \quad \gamma = \gamma_0 + (n_M - n_E)t
$$

$$
\begin{cases} \gamma_0 = \gamma_f + (n_M - n_E)t_{wait} \\ \gamma = \gamma_0' = -\gamma_f \end{cases} \quad \Rightarrow \quad -\gamma_f = \gamma_f + (n_M - n_E)t_{wait}
$$





2 Part 2

2.1 Problem

Then analyze the departure from LEO, verifying that it is convenient to use a departure from an orbit of minimum height (200 km), finding the sensitivity of the injection speed on the hyperbolic excess of speed and the geometry of the starting hyperbola. Evaluate the mass of the spacecraft heading towards Mars, assuming that the total mass in parking orbit (4000 kg) is accelerated with a solid propellant thruster (specific impulse $I_{sp} = 290\ s$, structural efficiency $\epsilon = 10$).

2.2 The optimal height to depart

Departure is usually from a parking orbit, the altitude of which derives from a compromise between gravitational losses (smaller, for lower orbits) and atmospheric drag (that are higher if r_{LEO} is low).

$$r_{LEO} = [100, \dots, 500] + r_e$$
$$V_{LEO} = \sqrt{\frac{\mu_E}{r_{LEO}}}$$
$$V_{hyp,per} = \sqrt{V^2_{\infty,dep} + \frac{2\mu_E}{r_{LEO}}}$$
$$V_{0,eq} = \sqrt{2\left(\frac{\mu_E}{r_E} - \frac{\mu_E}{2r_{LEO}}\right)}$$
$$\Delta V_{2,id} = V_{hyp,per} - V_{LEO}$$
$$\Delta V_2 = \Delta V_{2,id} + V_{0,eq}$$

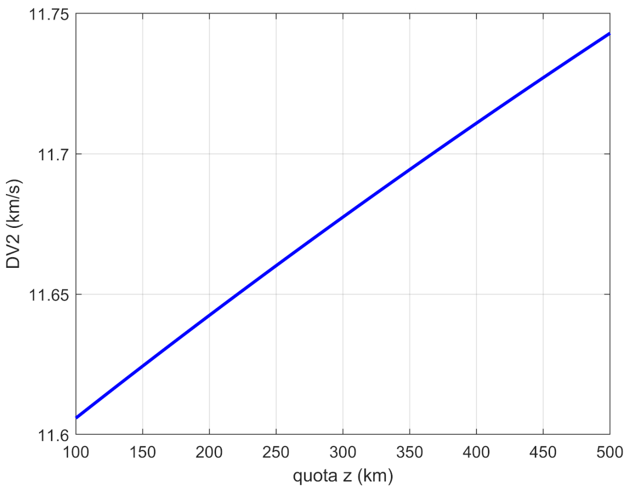


FIGURE 1. Injection cost ΔV_2 as height z varies

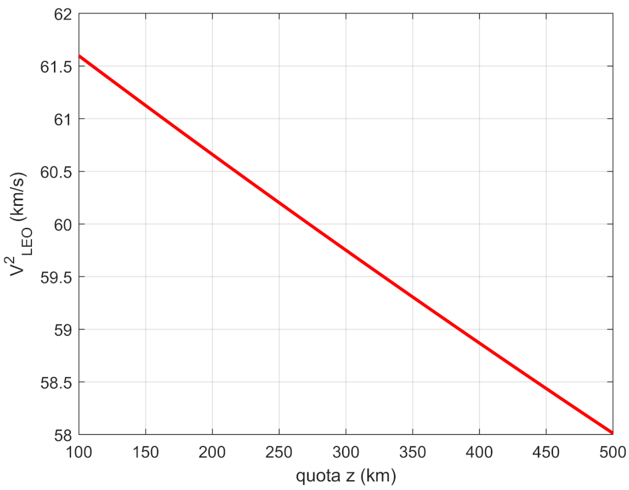


FIGURE 2. Drag $D \propto V^2_{LEO}$ as height z varies

2.3 Sensitivity of the injection speed

Injection errors must be kept as low as possible, as the hyperbolic excess speed depends strongly on the value of injection velocity, V_{LEO}.

$$f = \left(\frac{V_{hyp,per}}{V_{\infty,dep}}\right)^2 \approx 16$$
$$\frac{dV_0}{V_0} = \frac{1}{f}\frac{dV_\infty}{V_\infty}$$

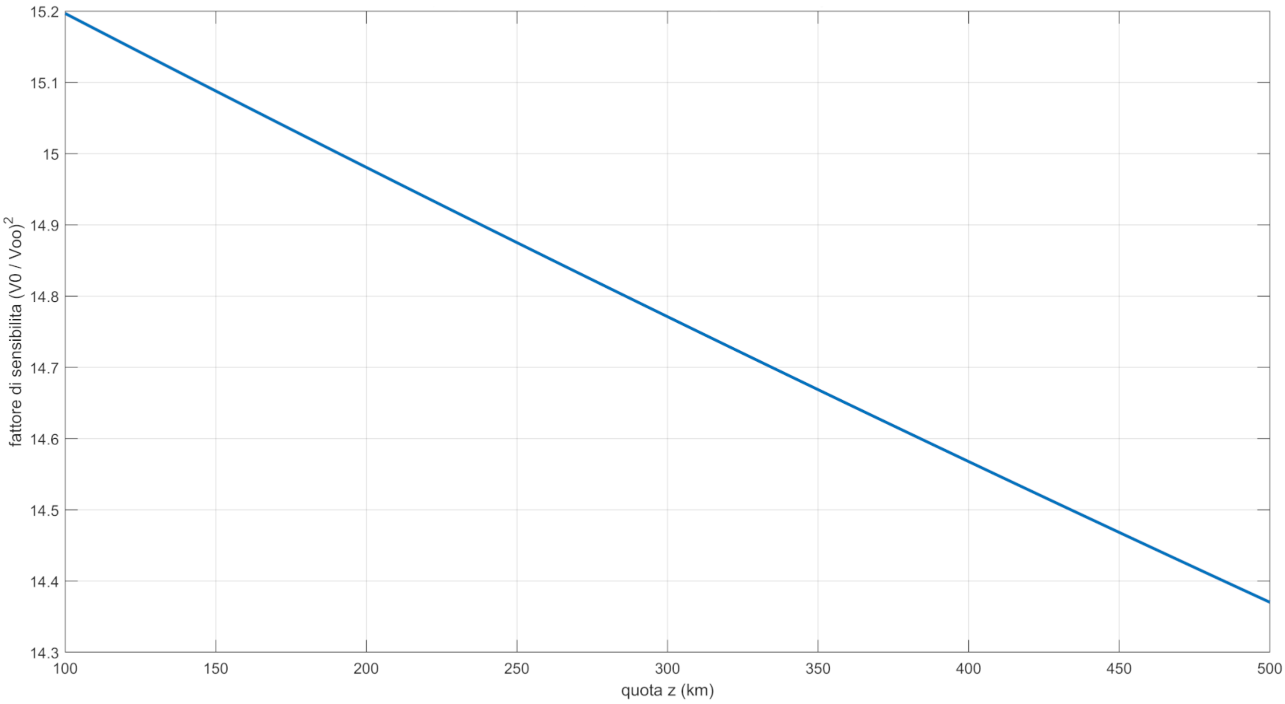


FIGURE 3. Sensitivity factor f as height z varies

2.4 The departure hyperbola

We have chosen a value of $z = 200\ km$, that meaning $r_{LEO} = 6571\ km$.

$$\epsilon_{hyp} = \frac{V_{\infty,dep}^2}{2} = 4.3389\ km^2/s^2$$
$$h_{hyp} = V_{hyp,per} r_{LEO} = 74921\ km^2/s$$
$$e_{hyp} = \sqrt{1 + 2\epsilon_{hyp}\frac{h_{hyp}^2}{\mu_E^2}} = 1.1431$$
$$\beta_{hyp} = \arccos\left(\frac{1}{e_{hyp}}\right) = 0.5057\ rad = 28.97°$$
$$a_{hyp} = -\frac{\mu_E}{2\epsilon_{hyp}} = -45934\ km$$

2.5 Estimation of the spacecraft mass

Inverting the *Tsiolkovsky's equation* we can deduce the mass of the spacecraft at the end of the hyperbolic injection and therefore how the mass is distributed.

$$
\begin{aligned}
&g_0 = 9.81 \times 10^{-3}\ km/s^2 \\
&M_0 = 4000\ kg \\
&\Delta V = g_0 I_{sp} \ln\left(\frac{M_0}{M_f}\right) \Rightarrow M_f = M_0 \exp\left(-\frac{\Delta V_{2,id}}{I_{sp} g_0}\right) = 1123\ kg \\
&M_p = M_0 - M_f = 2877\ kg \\
&M_s = \frac{M_p}{\epsilon - 1} = 319.6\ kg \\
&M_u = M_f - M_s = 803.6\ kg
\end{aligned}
$$

Subsystem	Mass [kg]	Margin	Mass total [kg]
Power	43.00	2.75	45.75
Payload	31.00	2.96	33.96
Communications	28.20	5.22	33.42
Onboard Data Handling / Avionics	15.00	0.75	15.75
AOCS	92.86	2.45	95.31
Thermal Control	24.00	1.40	35.20
Additional Shielding	42.67	0.00	42.67
Chemical Propulsion System (dry mass)	37.60	3.63	41.23
Harness (5%)	15.72	0.00	15.72
Structure (20% of dry mass)	66.01	0.00	66.01
TOTAL (dry, without system margin)	396.05	22.18	418.23
System Margin (20%)	79.21		
TOTAL (dry, with margin)			497.44
Propellant			306.2
TOTAL (wet mass)			803.64

FIGURE 4. Mass budget of a spacecraft

3 Part 3

3.1 Problem

Analyze the arrival on Mars in the following cases:

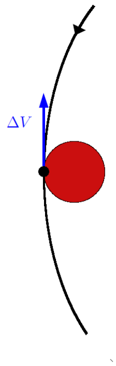
- soft landing
- 3597 km circular orbit
- circular orbit with period equal to 1 solar day (reached directly or with a two-pulse maneuver)
- same circular orbit reached with three-pulse maneuver (with the second at a radius of 100,000 km)
- elliptical orbit with period 1 solar day and pericenter equal to 3597 km

3.2 Solution

- A : soft landing

$$V_{hyp,per} = \sqrt{V_{\infty,arr}^2 + \frac{2\mu_M}{r_M}} = 5.68\ km/s$$

$$\Delta V = V_{hyp,per} = 5.68\ km/s$$



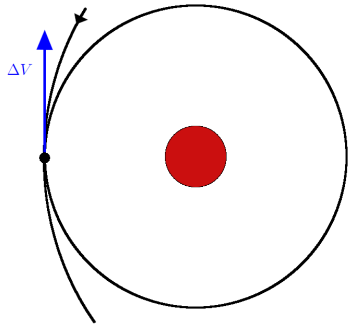
- B : 3597 km circular orbit

$$r_{per} = 3597\ km$$

$$V_{hyp,per} = \sqrt{V_{\infty,arr}^2 + \frac{2\mu_M}{r_{per}}} = 5.55\ km/s$$

$$V_{per} = \sqrt{\frac{\mu_M}{r_{per}}} = 3.45\ km/s$$

$$\Delta V = V_{hyp,per} - V_{per} = 2.10\ km/s$$



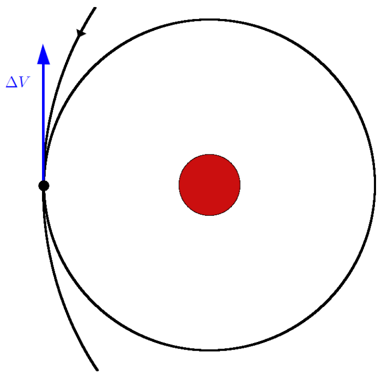
- C1 : circular orbit with period equal to 1 solar day (reached directly)

$$r = \sqrt[3]{\mu_M \left(\frac{T_M}{2\pi}\right)^2} = 20448\ km$$

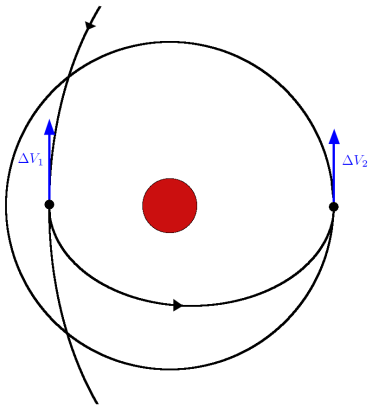
$$V_{hyp,per} = \sqrt{V_{\infty,arr}^2 + \frac{2\mu_M}{r}} = 3.3482\ km/s$$

$$V = \sqrt{\frac{\mu_M}{r}} = 1.45\ km/s$$

$$\Delta V = V_{hyp,per} - V = 1.9010\ km/s$$



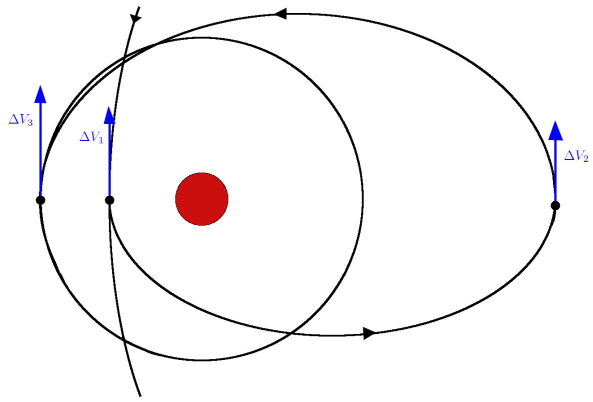
- C2 : circular orbit with period equal to 1 solar day (reached with a two-pulse maneuver)



$$r_{per} = 3597\ km$$
$$r_{apo} = 20448\ km$$
$$a = \frac{r_{per} + r_{apo}}{2} = 12022\ km$$
$$e = \frac{r_{apo} - r_{per}}{r_{apo} + r_{per}} = 0.70$$
$$V_{per} = \sqrt{\frac{\mu_M}{a}}\sqrt{\frac{1+e}{1-e}} = 4.5001\ km/s$$
$$V_{apo} = \sqrt{\frac{\mu_M}{a}}\sqrt{\frac{1-e}{1+e}} = 0.7916\ km/s$$

$$\Delta V_1 = V_{hyp,per} - Vper = 1.0528\ km/s$$
$$\Delta V_2 = V_{C1} - V_{apo} = 0.6556\ km/s$$
$$\Delta V = \Delta V_1 + \Delta V_2 = 1.7085\ km/s$$

- D : same circular orbit reached with three-pulse maneuver (with the second at a radius of 100,000 km)

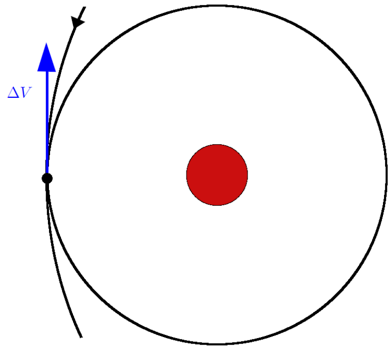


$$r_0 = 3597\ km$$
$$\bar{r} = 100000\ km$$
$$r_f = 20448\ km$$
$$a_1 = \frac{r_0 + \bar{r}}{2} = 51799\ km$$
$$e_1 = \frac{\bar{r} - r_0}{\bar{r} + r_0} = 0.9306$$
$$a_2 = \frac{\bar{r} + r_f}{2} = 60224\ km$$
$$e_2 = \frac{\bar{r} - r_f}{\bar{r} + r_f} = 0.6605$$
$$V_{ell,per,1} = \sqrt{\frac{\mu_M}{a_1}}\sqrt{\frac{1+e_1}{1-e_1}} = 4.7944\ km/s$$
$$V_{ell,apo,1} = \sqrt{\frac{\mu_M}{a_1}}\sqrt{\frac{1-e_1}{1+e_1}} = 0.1725\ km/s$$
$$V_{ell,per,2} = \sqrt{\frac{\mu_M}{a_2}}\sqrt{\frac{1+e_2}{1-e_2}} = 1.8649\ km/s$$
$$V_{ell,apo,2} = \sqrt{\frac{\mu_M}{a_2}}\sqrt{\frac{1-e_2}{1+e_2}} = 0.3813\ km/s$$

$$\Delta V_1 = V_{hyp,per,B} - V_{ell,per,1} = 0.7585\ km/s$$
$$\Delta V_2 = V_{ell,apo,2} - V_{ell,apo,1} = 0.2089\ km/s$$
$$\Delta V_3 = V_{ell,per,2} - V_{C1} = 0.4177\ km/s$$
$$\Delta V = \Delta V_1 + \Delta V_2 + \Delta V_3 = 1.3851\ km/s$$

- E : elliptical orbit with period 1 solar day and pericenter equal to 3597 km

$$a = r_{C1} = 20448\ km$$
$$r_{per} = r_B = 3597\ km$$
$$r_{apo} = 2a - r_{per} = 37299\ km$$
$$e = \frac{r_{apo} - r_{per}}{r_{apo} + r_{per}} = 0.8241$$
$$V_{ell,per} = 4.6603\ km/s$$
$$\Delta V = V_{hyp,per,B} - V_{ell,per} = 0.8926\ km/s$$



4 Part 4

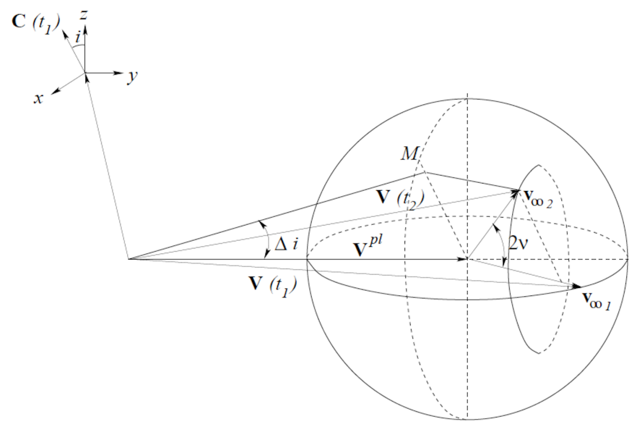
4.1 Problem

Wanting to use the flyby of Mars to rotate the plane of the heliocentric trajectory, calculate the inclination obtainable if the pericenter of the hyperbola is 200 km above the planet's surface.

4.2 Plane rotation due to the flyby

A gravity assist maneuver at an attracting body can be used to implement a controlled change of the spacecraft characteristics: here we analyze the effect on inclination. The highest potentialities for changing the spacecraft orbital plane in a single fly-by is typical of those large planets of the Jovian group (Jupiter, Saturn, Uranus and Neptune). The potentialities of the Earth group planets (Earth, Venus, Mars and Mercury) are somewhat lower.

$$r_{per} = 200 + r_M$$
$$\delta = 2 \arcsin\left(\frac{1}{1 + \frac{r_{per} V_\infty^2}{\mu_M}}\right) = 77.96°$$
$$\Delta i = \arcsin\left(V_{\infty,arr}^2 \frac{\sin\delta}{V_M}\right) = 6.164°$$
$$\Delta i_{max} = \arcsin\left(\frac{V_{\infty,arr}^2}{V_M}\right) = 6.303°$$



5 Part 5

5.1 Problem

Finally, consider the time interval (launch window) in which the mission is possible by reducing the mass of the spacecraft by 1%. Draw the heliocentric trajectories at the opening and closing of the window in an inertial reference.

5.2 Removal of 1% of M_u

We reduce here the mass of the spacecraft allocated to the probe, while retaining the total mass. Due to the change, we need to estimate the new distribution of mass.

$$
\begin{aligned}
&{M_u}' = 0.99 \cdot M_u = 795.6\ kg \\
&dM_p = \frac{0.01 \cdot M_u}{1 + \frac{1}{\epsilon - 1}} = 7.23\ kg \\
&dM_s = 0.01 \cdot M_u - dM_p = 0.80\ kg \\
&{M_p}' = M_p + dM_p = 2884\ kg \\
&{M_s}' = M_s + dM_s = 320.4\ kg \\
&{M_0}' = {M_u}' + {M_s}' + {M_p}' = 4000\ kg \\
&{M_f}' = {M_u}' + {M_s}' = 1116\ kg
\end{aligned}
$$

5.3 Estimation of launch window

To estimate the launch window length we need to identify its limits: that means the first moment we can depart to Mars and the last possible moment to do so. These time instants will be called $\gamma(1)$ and $\gamma(2)$.

$$
\begin{aligned}
&\Delta V = I_{sp} g_0 \ln\left(\frac{{M_0}'}{{M_f}'}\right) = 3.6316\ km/s \\
&V_{hyp,per} = \Delta V + \sqrt{\frac{\mu_E}{r_{LEO}}} \\
&V_{\infty,dep} = \sqrt{V_{hyp,per}^2 - \frac{2\mu_E}{r_{LEO}}} \\
&V_{dep} = V_E + V_{\infty,dep} \\
&h = V_{dep} R_E \\
&p = \frac{h^2}{\mu_S} \\
&e = \frac{p}{R_E} - 1 \\
&a = \frac{p}{1 - e^2} \\
&\nu = \begin{cases} \arccos\left(\frac{p - R_M}{R_M e}\right) \\ 2\pi - \arccos\left(\frac{p - R_M}{R_M e}\right) \end{cases} \\
&E = 2 \arctan\left(\sqrt{\frac{1-e}{1+e}} \tan\frac{\nu}{2}\right)
\end{aligned}
$$

$$TOF = \sqrt{\frac{a^3}{\mu_S}} \cdot \Delta M$$
$$\gamma = \nu - TOF \cdot n_M$$
$$\Delta t = \frac{\gamma(1) - \gamma(2)}{n_E - n_M} \cdot \frac{1}{86400} = 9.7160\ days$$

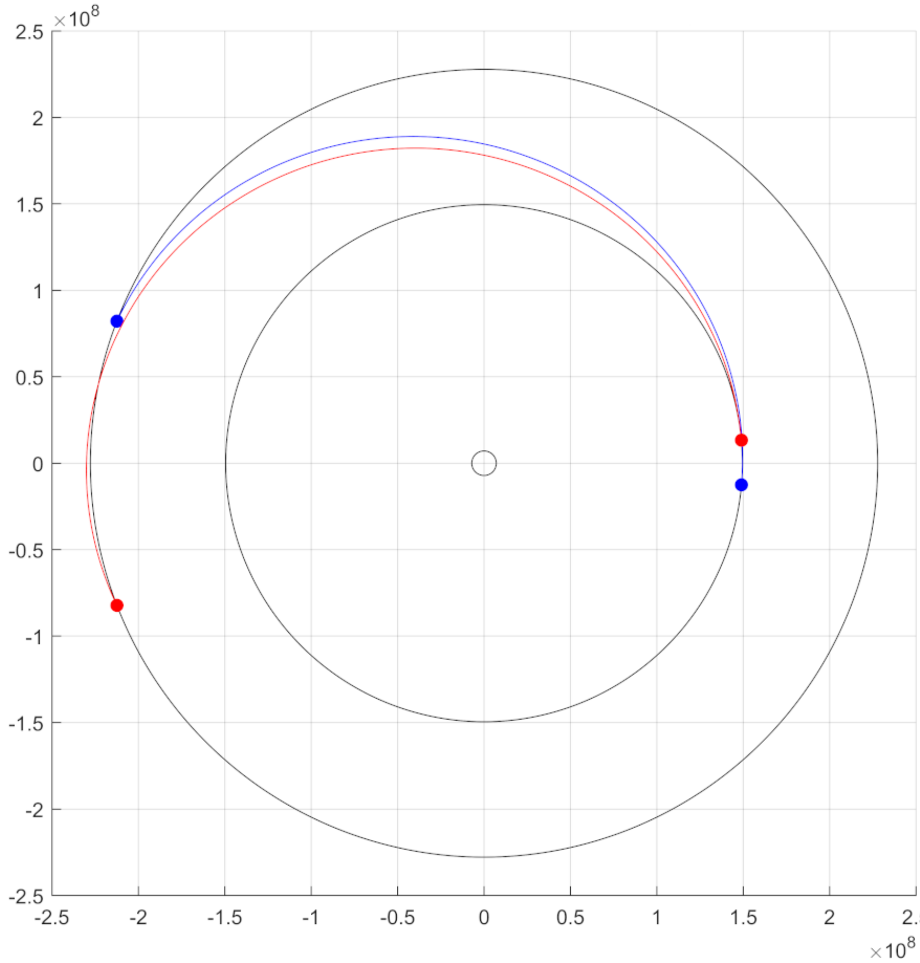


FIGURE 5. Launch window for departure for the Earth-Mars mission

In Fig. 4 we can see the launch window, when it starts (in blue) and when it ends (in red). This allows us to be a little more relaxed on the requirements.